

Portfolio Six: Lists

Program: Cat food Inventory

This program tracks, measures, and organize cat food stock so a cat shelter or store never runs out of food.

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In [16]: # Creating a list collection to hold cat food item weights in grams
food_bags = [430, 1300, 400, 300, 500]

# Changing the weight of the third bag (spot 2) because a cat ate some
print("Old weight at spot 2:", food_bags[2])
food_bags[2] = 200
print("New weight at spot 2:", food_bags[2])

# append() adds a new food bag and remove() deletes an empty food bag.
food_bags.append(1000)
print("Inventory after buying a bag:", food_bags)

food_bags.remove(200)
print("Inventory after removing empty a bag:", food_bags)

total_bags = len(food_bags)    # Counts the the food bags
smallest_bag = min(food_bags)  # Finds the smallest food bag
largest_bag = max(food_bags)   # Finds the largest food bag
total_weght = sum(food_bags)   # Adds all food weights

# Arranging the foods from the light to heavy
food_bags.sort()
print("Food stock sorted from low to high:", food_bags)

# Pointing directly to the last item to find the heaviest bag of food
heaviest_bag = food_bags[-1]
print("The single heaviest food bag is:", heaviest_bag)

# Looking at every single food bag in the inventory one by one
print("---- Scanning Food Stock ----")
for weight in food_bags:
    print("Found cat food weighing:", weight, "grams")

# Splitting a text line of food flavors into lists of words
flavor_text = "Salmon,Chicken,Tuna,Beef,Mackerel"
flavor_list = flavor_text.split(",")

print("Raw text line of single flavors:", flavor_text)
print("Clean list of single flavors:", flavor_list)

# Going through the new flavor list
for flavor in flavor_list:
    print("Available food flavor logged:", flavor)
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Old weight at spot 2: 400
New weight at spot 2: 200
Inventory after buying a bag: [430, 1300, 200, 300, 500, 1000]
Inventory after removing empty a bag: [430, 1300, 300, 500, 1000]
Food stock sorted from low to high: [300, 430, 500, 1000, 1300]
The single heaviest food bag is: 1300
---- Scanning Food Stock ----
Found cat food weighing: 300 grams
Found cat food weighing: 430 grams
Found cat food weighing: 500 grams
Found cat food weighing: 1000 grams
Found cat food weighing: 1300 grams
Raw text line of single flavors: Salmon,Chicken,Tuna,Beef,Mackerel
Clean list of single flavors: ['Salmon', 'Chicken', 'Tuna', 'Beef', 'Mackerel']
Available food flavor logged: Salmon
Available food flavor logged: Chicken
Available food flavor logged: Tuna
Available food flavor logged: Beef
Available food flavor logged: Mackerel

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In [7]: ## Concepts that were used here:
# Concept of a collection
# Lists and definite loops
# Indexing and lookup
# List mutability
# Functions
# Slicing lists
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# List methods
# Sorting lists
# Splitting strings into words
# Using split to parse strings
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